CHAPTER 1

INTRODUCTION

This chapter explain about project motivation and specific details of the Centralized Web-Based System for Stress Induction Methods. This chapter start with the project background and problem statement, then followed by the aim and goals. The Centralized Web-Based System for Stress Induction Methods significance, scopes, thesis outline, and project summary are also included in this chapter.

1.1 Background of Study

Psychological stress can be explained as a kind of psychological tension or strain which is hard to cope with or sustain. In the human brain, stress can be equated as something that has an influence through emotions, negative thought, or other behaviour that can cause an effect on the physiology as well as the psychology of an individual (Allen et al., 2017). From Malaysia Youth Mental Health Index 2023, from 2022 showed that 1 in 10 youth experience high stress (UNICEF, 2024). The index giving the rising awareness of youth stress.

Several tests have been developed for this purpose. Among the most widely used standardized stress induction tests in psychological research are the Stroop Test, Montreal Imaging Stress Task (MIST), and Trier Social Stress Test (TSST) (Kirschbaum et al., 1993; Dedovic et al., 2005; Dickerson & Kemeny, 2004). The Stroop Test, also known as the colour-word naming test has been extensively used in psychological research since it was first described by Stroop (Jensen, 1965). The test has been used to assess a range of processes and characteristics in the cognitive and personality domains in more than sixty published psychological research (Jensen, 1965). Next, the Montreal Imaging Stress Task (MIST) combines computerized mental arithmetic challenges with social evaluative threat components built into the program or presented by the investigator (Dedovic et al., 2005). The three phase in MIST are experimental, control, and rest. During the rest phase, participant will be shown nothing as the starting point of the test. The control condition involves the participant must

answer the arithmetic question in easy level while experimental phase is the one that pressure the participant where it is giving the almost impossible arithmetic task. Additionally, performance feedback is shown to boost stress and social assessment (Dedovic et al., 2005). Meanwhile, the Trier Social Stress Test (TSST) is an ecologically sound technique that uses social assessment and uncertainty to create acute psychological stress. TSST is mostly dependent on stress caused by public speaking and the mental performance being observed (Allen et al., 2017).

An interview had been conducted with Mr. Muhammad Hamiz bin Mohd Radzi, a stress researcher, which specify that in practical research settings, these stress induction methods are often conducted using multiple independent tools and platforms. Based on the interview also, the Stroop Test is currently conducted using specialized experimental software such as E-Prime, while instructions for the TSST are delivered using presentation slides, and timing is sometimes controlled using manual tools such as a stopwatch. In addition, some participant responses, particularly during arithmetic tasks, are still recorded manually by observers. This fragmented setup creates difficulties in maintaining experimental flow and increases the complexity of managing research data.

Examining stress in daily life is made possible by digital technology, but majority of standardized stress induction techniques are still dispersed across multiple platform (Vikoler et al., 2025). Researchers have demanded a verified and standardized stress induction technique that can be used digitally and is not limited by laboratory conditions to overcome these difficulties (Norden et al., 2022).

1.2 Problem Statement

The following provides more detail on each of these issues:

a) Lack of centralized platform combining multiple stress tests.

According to the interview conducted, the experiment tools are disconnected as the researchers must switch between different programs during a single

experimental session. This disrupts the experimental flow and may interrupt the stress level experienced by participants, which can negatively affect the validity of the collected data. This is aligned as cited by Norden et al (2022) which stated that, the Stroop Test, TSST, and MIST are conducted in different websites, software, or manual setups.

b) The recorded reaction result for each test is not easily available.

Another major issue is the difficulty in managing and synchronizing participant data collected during stress induction experiments. Based on the interview, while some computerized tests automatically store reaction time data, other tasks such as arithmetic performance in TSST require manual recording by observers. These manual recordings are often written on paper and later transferred to digital formats such as spreadsheets. Manual recording increases the risk of transcription errors (Ab Rahman et al., 2025), inconsistent data formats, and difficulties in merging datasets from different tools (Becerra et al., 2025).

1.3 Proposed Solution

To address the problem stated in the problem statement, this project proposes the development of a Centralized Web-Based System for Stress Induction Methods which have two main approaches:

a) Centralized Integration of Stress Tests: To resolve the lack of a centralized platform, the project involves in developing centralized web-based system which combine multiple stress test, which is Stroop Test, MIST, and TSST Arithmetic Task into one accessible environment.
b) Automated Behavioural Recording: To address the difficulty in accurately capturing and accessing reaction result, the system will feature an integrated video recording function. This will automatically record participants reaction during the stress test.

1.4 Project Aim

This project aims to develop a centralized web-based system that combines multiple stress induction methods, including the Stroop Test, MIST, and TSST Arithmetic Task, on one website, offering integrated platforms for all users.

1.5 Project Objectives

The three objectives of this project are:

a) To gather and analyse requirements for Centralized Web-Based System for Stress Induction Methods.
b) To design the Web-Based System for Stress Induction Methods.
c) To develop a Centralized Web-Based System that integrates the Stroop Test, MIST, and TSST Arithmetic Task.

1.6 Project Scopes

This project focuses on developing a centralized web-based system that integrates multiple stress induction methods include Stroop Test, MIST, and TSST Arithmetic Task into a single centralized web-based platform making it able to use for researchers who need to do the test. The system features will include video recording integration for participant reaction observation and minimizing human error by using traditional way. Additionally, the system aims is to solve issue of scattered website and manual setup for stress induction methods, making it convenient for researcher to use for their research or monitor their patients.

The user for this system is for researchers and their participants; however, other users such as students can still use it for experimental purposes. This system will be developed by using PHP language and implementing MariaDB for database.

1.7 Project Significances

The project significance is the proposed system will provide a centralized platform that integrates multiple stress induction methods such as the Stroop Test, MIST, and TSST Arithmetic Task into a single web-based environment. Based on feedback obtained from the stakeholder interview, researchers currently spend significant time switching between different software tools and manually compiling experimental data. The centralized system will reduce these inefficiencies by providing automated data collection, synchronized timing, and integrated recording features. This can significantly reduce manual

workload, minimize human error, and improve the consistency of experimental procedures.

1.8 Outline of the Thesis

There are five chapters in this thesis which are:

Table 1.1 Chapter in This Proposal

Chapter 2	This chapter discusses the literature review that related to The Centralized Web-Based System for Stress Induction Methods. The previous research and article are used to identify the problem and improvements needed to achieve. This chapter also show the description of three methodologies that fall under the software development life cycle (SDLC).
Chapter 3	In this chapter, the methodology planned to be used in development of The Centralized Web-Based System for Stress Induction Methods is elaborated. It includes the system development model, tools, and language used such as PHP.
Chapter 4	This chapter shows the results of work done for each phase described in the methodology chapter. Thorough analysis and discussions of each finding are described here.
Chapter 5	This chapter concludes the research with discussion of the benefits gained with the system and recommendation for future works.

1.9 Summary

This chapter has discussed the context of the study, which includes the setting, the problem statement on the project of the Centralized Web-Based System for Stress Induction Methods, the purpose, and the aim of the project. This concludes the background chapters of the study.